

CHAPTER 04 Metals

At the end of this chapter, you will be able to describe the properties and uses of metals. You will see that the properties of metallic elements differ from those of non-metals. You will also see how you can utilise the properties, such as malleability, thermal conductivity, high melting point and electrical conductivity in a range of everyday applications.

You will learn how chemists have been able to relate these properties to the structure of metals and be able to explain their structure in terms of a metallic bonding model. Your study will enable you to understand that chemists model metals as lattices of positive ions held together by the electrostatic force that exists between these ions and the delocalised outer shell electrons. Your previous study of chemical bonding allows you to explain that bonding occurs so that electron arrangements in atoms change in order to stabilise the valence shell.

Science as a Human Endeavour

- Matter at the nanoscale can be manipulated to create new materials, composites and devices; the different characteristics of nanomaterials can be used to provide commercially available products. As products are designed on the basis of properties which are different from the bulk material, their use can be associated with potential risks to health, safety and the environment and this has led to regulations being developed to address new and existing nanoform materials.

Science understanding

- metallic bonding can be modelled as a regular arrangement of atoms with electrostatic forces of attraction between the nuclei of these atoms and their delocalised electrons that are able to move within the three-dimensional lattice
- the metallic bonding model can be used to explain the properties of metals, including malleability, thermal conductivity, generally high melting point and electrical conductivity; covalent bonding can be modelled as the sharing of pairs of electrons resulting in electrostatic forces of attraction between the shared electrons and the nuclei of adjacent atoms

4.1 Properties of metals

More than 80% of the elements in the periodic table are metals.

Metals have been important to human beings since early times. The development of civilisation can be measured by the way we have used metals. The Copper Age (5000–3000 BCE) was followed by the Bronze Age (3000–1000 BCE) and the Iron Age (from 1000 BCE).

Gold, silver and copper can be found on Earth in an almost pure form. These metals were employed by prehistoric humans to make ornaments, tools and weapons. As humans' knowledge of metallurgy (the science of modifying metals) developed, metals have played a central role in fields as diverse as construction, agriculture, art, medicine and transport.

The diverse properties of different metals make them suitable for many purposes. Table 4.1.1 shows the properties and uses of some metals. For example, titanium (Figure 4.1.1) is a very strong, relatively unreactive metal with a low density that is close to that of bone. Consequently, it is used in surgical implants that can last up to 20 years with little effect on the body. Titanium is also used in the aerospace industry, in art and architecture, and in sporting products such as golf clubs.

TABLE 4.1.1 Properties and uses of some metals

Metal	Properties	Uses
iron	soft, malleable, magnetic, good thermal and electrical conductor, fairly reactive, readily forms alloys	can corrode and is usually converted to more stable steel, which is used in buildings and bridges, automobiles, machinery and appliances
aluminium	low density, relatively soft when pure, excellent thermal and electrical conductor, malleable and ductile, good reflector of heat and light, readily forms alloys	saucepans, frying pans, drink cans, cooking foil, food packaging, roofing, window frames, appliance trim, decorative furniture, electrical cables, aircraft and boat construction
titanium	very strong, high melting point, low density, low reactivity, readily forms alloys	medical devices within the body, wheelchairs, computer cases; lightweight alloys are used in high-temperature environments such as spacecraft and aircraft
gold	shiny gold appearance, excellent thermal and electrical conductor, unreactive, readily forms alloys	electrical connections, jewellery, monetary standard, dentistry



FIGURE 4.1.1 Titanium has many uses: (a) a replacement hip socket, (b) the spectacular curved space museum building in Moscow and (c) the SR-71 Blackbird reconnaissance aircraft. The SR-71 aircraft is the fastest manned aircraft that uses oxygen directly from the atmosphere.

In this section, you will examine the properties of metals. Then, in the next section, you will learn about the bonding model that chemists have developed to explain these properties. This model has helped chemists and materials engineers to understand why metals behave the way they do and how metals can be modified to create useful new materials.

PROPERTIES OF METALS

Table 4.1.2 gives the properties of some metals and non-metals. Despite the different properties of metals, most metals:

- exhibit a range of melting points and relatively high boiling points
- are good **conductors** of electricity
- are good conductors of heat
- generally have high densities.

TABLE 4.1.2 Properties of some metallic and non-metallic elements

Element	Melting point (°C)	Boiling point (°C)	Electrical conductivity (MS m ⁻¹)*	Thermal conductivity (Js ⁻¹ m ⁻¹ K ⁻¹)†	Density (gmL ⁻¹)
Metals					
gold	1063	2970	45	310	19.3
iron	1540	3000	9.6	78	7.86
mercury	-39	357	1	8.4	13.5
potassium	64	760	14	100	0.86
silver	961	2210	60	418	10.5
sodium	98	892	21	135	0.97
Non-metals					
carbon (diamond)	3550	‡	10 ⁻¹⁷	–	3.51
oxygen	-219	183	–	0.026	1.15 (liquid)

*MS m⁻¹ = megasiemens per metre.

†Thermal conductivity measures the conductance of heat.

‡Diamond sublimates (changes straight from a solid to a gas) when heated.

Not all metals have all of these properties. Mercury is a liquid at room temperature—it has an unusually low melting point. The group 1 elements (the **alkali metals**) have some properties that make them different from most other metals. They are all soft enough to be cut with a knife and they react vigorously with water to produce hydrogen gas. Both mercury and the group 1 elements exhibit most of the other properties listed above and are classified as metals.

Metals also generally have the following characteristics in common.

- They are **malleable**—they can be shaped by beating or rolling.
- They are **ductile**—they can be drawn into a wire.
- They are lustrous or reflective when freshly cut or polished.
- They are often hard, with high **tensile strength** (the force required to pull a material to the point where it breaks).
- They have low ionisation energies (a measure of the energy required to remove an electron from a gaseous atom) and electronegativities (a measure of the tendency of an atom to attract a bonding pair of electrons).

These properties can allow different metals to be used together in order to solve many engineering problems. The power transmission tower in Figure 4.1.2 is made of a few metals to take advantage of their different properties.



FIGURE 4.1.2 This power transmission tower relies on the strength of iron in steel for its structural integrity. The electricity cables are made from aluminium, utilising its ductility and electrical conductivity.

i Most metals are good conductors of electricity and heat. They are malleable, ductile, and have high tensile strength and low ionisation energies.

Generally, metals are shaped for use in different applications by hammering, exploiting their malleability. Some metals, such as gold, copper and aluminium, are very malleable at room temperature. Other metals, such as iron, must be heated before they can be shaped.

Most metals are similar in appearance, being lustrous (reflective) and silvery-grey. Gold and copper are notable exceptions. Gold is a yellow coloured metal; copper is reddish.

CHEMFILE

Beryllium

Beryllium is the fourth element in the periodic table. It is one of the lightest metals, its density being two-thirds the density of aluminium. Beryllium is non-magnetic and has six times the stiffness of steel. The Space Shuttle and the Spitzer Space Telescope both use beryllium due to its strength and light weight. NASA's next-generation James Webb Space Telescope shown in Figure 4.1.3, scheduled for launch in 2018, will depend on a 6.5-metre mirror constructed using beryllium to see objects 200 times fainter than those previously visible.

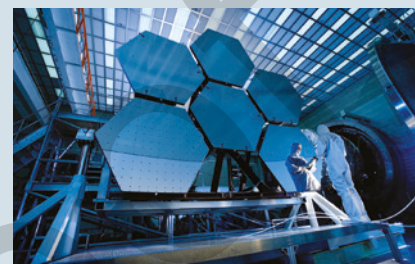


FIGURE 4.1.3 Some of the 18 mirror segments of the James Webb Space Telescope. The mirrors are supported by beryllium ribs that maintain the mirror's shape under extreme conditions.

FORMING METAL IONS

In Chapter 3, you saw that there is a general tendency for atoms to combine so that they have eight electrons in the outer shell. This is called the octet rule.

Metallic elements are found on the left-hand side of the periodic table. The atoms of metals are generally larger than the atoms of non-metallic elements within a period and the core charge of their atoms is lower. The core charge of an atom is a measure of the attractive force felt by the outer-shell electrons towards the nucleus. It takes less energy to remove electrons from an outer shell when an atom is large, so the ionisation energy of metals is usually lower than for non-metals in the same period. As a consequence, metal atoms tend to lose their outer-shell electrons to form positive ions, called cations.

Atoms of simple metals have one, two or three electrons in their outer shell. The cations that are formed when these metal atoms lose these valence electrons have a stable noble gas electron configuration, with eight electrons in their outer shell.

i Metals tend to lose valence electrons and form cations.

Worked example 4.1.1

DETERMINING CHARGES

Determine the charge of a calcium cation.	
Thinking	Working
Unreacted calcium atoms have the same number of protons and electrons.	Atomic number (Z) of calcium is 20: number of protons is 20, number of electrons is 20
The electrons in an atom are in shells.	Shell configuration of calcium: 2,8,8,2
Only the outer-shell electrons will be lost.	Outer shell contains two electrons, $20 - 2 = 18$ electrons remaining
Cation charge = number of protons – number of electrons	Cation charge = $20 - 18 = +2$

Worked example: Try yourself 4.1.1

DETERMINING CHARGES

Determine the charge of an aluminium cation.

TRANSITION METALS

Between group 2 and group 13 in the periodic table is a block of elements known as the transition metals (Figure 4.1.4.). They include metals such as iron and nickel that are used to build cities, bridges, cars and railway lines, and precious metals such as silver and gold that have ornamental and economic uses. Most transition metals are silver-coloured and are similar in appearance, as can be seen in Figure 4.1.5.

The figure shows a standard periodic table. The transition metal block, located between groups 2 and 13, is highlighted in blue. The label 'FPO' is superimposed over the center of this block. The elements in the highlighted block are Scandium (Sc), Titanium (Ti), Vanadium (V), Chromium (Cr), Manganese (Mn), Iron (Fe), Cobalt (Co), Nickel (Ni), Copper (Cu), and Zinc (Zn) in the first row; and Yttrium (Y), Zirconium (Zr), Niobium (Nb), Molybdenum (Mo), Technetium (Tc), Ruthenium (Ru), Rhodium (Rh), Palladium (Pd), Silver (Ag), and Cadmium (Cd) in the second row.

FIGURE 4.1.4 The transition metals are situated in the centre of the periodic table.

The transition metals are very important to Australian industry. All of the metals in the first row are found in Australia and many are being mined today.

Iron is by far the most important metal to us. Nearly ten times more iron is mined than all other metals combined. Iron obtained directly from a blast furnace is relatively **brittle** (meaning it snaps easily) and corrodes easily. Carbon and other transition metals are combined with iron to produce mixtures or alloys, called **steel**. Steel has more desirable characteristics than pure iron. (Alloys are covered in more detail in Section 4.4.)

Copper is one of the few transition metals that is mainly used in its pure form. It is highly electrical conductive and so it is used for most of the millions of kilometres of electrical wires that enable the transmission of electric energy for heating, lighting, telephones, radio and television.

Transition metals are not only important for industry, they are also important for life. All the transition metals in the first row, except scandium and titanium, are essential for animal life. Your body relies on the presence of trace elements to carry out certain biochemical reactions. For example, chromium, which you get from meat and bread, assists in the production of energy from glucose.

Properties of transition metals

Compared to the main group metals, transition metals have the following properties.

- They tend to be harder.
- They have higher densities.
- They have higher melting points.
- Some of them have strong magnetic properties.

The hardness, higher densities and higher melting points are due to the atoms of transition metals generally being a smaller size due to their greater core charge. This allows them to pack together more tightly with stronger bonds.

The high tensile strength of transition metals makes them suitable for use in the construction of buildings, cars, bridges and numerous other objects.



FIGURE 4.1.5 The first row of transition metals

EXTENSION

Transition metal compounds

Transition metal compounds display a wide range of different colours. They are extensively used as pigments in paints, and to colour glass, ceramics and enamel. In Figure 4.1.6, the colours used by the artists are caused by the different transition metals present and the colours are still as vivid today as when they were painted.

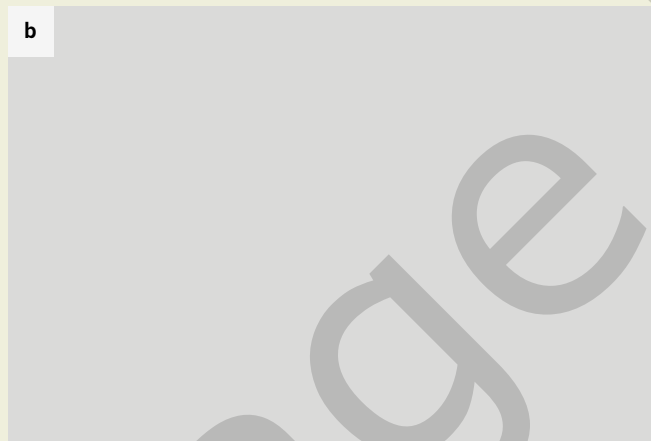


FIGURE 4.1.6 (a) Arthur Streeton's *Golden summer, Eaglemont*, 1889. (b) An Australian Aboriginal painting

Ochre is a type of hard clay that contains iron oxides and hydroxides, which can be found naturally in many colours, including red, pink, white and yellow. Ground into a powder and mixed with liquids, ochre forms a paste that has been used for millennia by Aboriginal and Torres Strait Islander people in Australia for body decoration, cave painting, bark painting and other artwork.

The colours of many gemstones are also due to the presence of transition metals. For example, sapphires (Figure 4.1.7) contain traces of titanium and iron in a crystal lattice of aluminium oxide. The particles in a crystal lattice are arranged in ordered and symmetrical arrangements, which are repeated at regular intervals.



FIGURE 4.1.7 Blue sapphires get their colour from impurities of titanium and iron.

The colours arise when electrons within the metal ions in the compounds absorb light of particular wavelengths and move to higher energy levels. Absorbance of light with some wavelengths and transmission of light with other wavelengths results in the compounds appearing coloured. By contrast, compounds of group 1 and group 2 metals are usually colourless.

4.1 Review

SUMMARY

- Metals have the following characteristic properties:
 - high boiling points
 - good conductors of electricity in solid and liquid states
 - malleable and ductile
 - high densities
 - good conductors of heat
 - lustrous
 - low electronegativities
 - low ionisation energies
 - react by losing electrons.
- The main differences between the properties of main group and transition metals are:
 - transition metals are harder
 - transition metals are more dense
 - transition metals have higher melting points
 - some transition metals have strong magnetic properties
 - transition metal compounds tend to be brightly coloured.

KEY QUESTIONS

- Determine the charge of the cations formed from the following metals if they lost all of their outer-shell electrons.
 - Li
 - Mg
 - Al
- Potassium is classed as a metal. Which of its properties are similar to those of the metal gold? In what ways is it different?
 - Identify another element in Table 4.1.2 on page xxx that has similar properties to potassium.
 - Identify another metal in Table 4.1.2 on page xxx that has similar properties to gold.
 - Where are these four metals in the periodic table?
- Which metals would you select if you wanted a good electrical conductor?
 - What other factors might influence your choice?
- Sodium and iron have very different physical properties. Explain why this is based on where these metals are found in the periodic table.
- Suggest some properties not included in Table 4.1.2 on page xxx that you would need to consider before choosing between aluminium and iron for building a bridge.

4.2 Metallic bonding

In this section, you will learn how the properties of metals can be explained in terms of the structure of the particles in metals. You will also learn about the bonding model that chemists have developed to explain these properties. The metallic bonding model has helped chemists and materials engineers to understand why metals behave as they do and how metals can be modified to create useful new materials.

CONNECTING PROPERTIES AND STRUCTURE

The properties of metals are listed in Table 4.2.1. Each of these properties gives some information about the structure and bonding of particles in metals.

TABLE 4.2.1 The physical properties of metals and resulting conclusions about metal structure and bonding

Property	What this tells us about structure
Metals are usually hard and tend to have high boiling points.	The forces between the particles must be strong.
Metals conduct electricity in the solid state and in the molten (melted) liquid state.	Metals have charged particles that are free to move.
Metals are malleable and ductile.	The attractive forces between the particles must be stronger than the repulsive forces between the particles when the layers of particles are moved.
Metals generally have high densities.	The particles are closely packed in a metal.
Metals are good conductors of heat.	There must be a way of quickly transferring energy throughout a metal object.
Metals are lustrous or reflective.	Free electrons are present, so metals can reflect light and appear shiny.
Metals tend to react by losing electrons.	Electrons must be relatively easily removed from metal atoms.

Chemists have developed a model for the structure of metals called the **metallic bonding model** that explains all the properties mentioned so far. You can deduce from the information in Table 4.2.1 that the metallic bonding model must include:

- charged particles that are free to move and conduct electricity
- strong forces of attraction between atoms throughout the metal structure
- some electrons that are relatively easily removed.

METALLIC BONDING MODEL

Electrons are the particles that enable metals to conduct electricity. They are able to move within the **lattice** (tightly packed arrangement) of metal atoms. Negatively charged electrons can be lost from the outer shell of metal atoms, forming positive ions (cations). As shown in Figure 4.2.1, the freed electrons **delocalise** (spread through a large area) to form a ‘sea’ of electrons throughout the entire metal structure and are strongly attracted to the metal cations.

Chemists believe that, in a solid sample of a metal:

- positive ions are arranged in a closely packed structure. This structure is described as a regular, three-dimensional network of positive ions. The cations occupy fixed positions in the lattice
- negatively charged electrons move freely throughout the lattice. These electrons are called **delocalised electrons** because they belong to the lattice as a whole, rather than staying in the shell of a particular atom

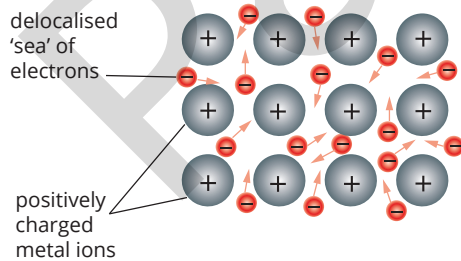


FIGURE 4.2.1 The metallic bonding model. Positive metal cations are surrounded by a mobile sea of delocalised electrons. This diagram shows just one layer of metal ions.

- the delocalised electrons come from the outer shells of the atoms. Inner-shell electrons are not free to move throughout the lattice and remain firmly bonded to individual cations
- the positive cations are held in the lattice by the electrostatic force of attraction between these cations and the delocalised electrons. This attraction extends throughout the lattice and is called **metallic bonding**.

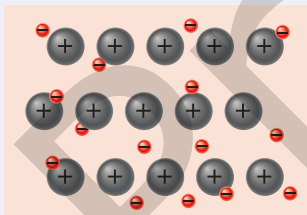
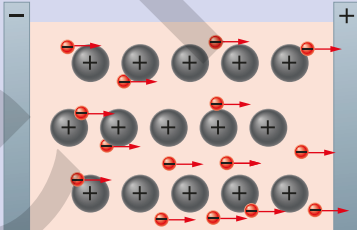
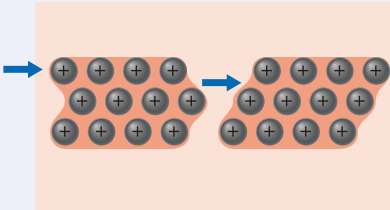
Together, these ideas make up the metallic bonding model. An example of how a metal, such as sodium, could be represented using this model is shown in Figure 4.2.2.

i In the metallic bonding model, positive metal cations are surrounded by a sea of delocalised electrons.

EXPLAINING THE PROPERTIES OF METALS

Table 4.2.2 shows how the metallic bonding model is consistent with the relatively high boiling point, electrical conductivity, malleability and ductility of metals.

TABLE 4.2.2 Physical properties of metals and explanations from metallic bonding model

Property	Explanation	
Metals are hard and have relatively high boiling points.	Strong electrostatic forces of attraction between positive metal ions and the sea of delocalised electrons hold the metallic lattice together.	
Metals are good conductors of electricity.	Free-moving delocalised electrons will move towards a positive electrode and away from a negative electrode in an electric circuit.	
Metals are malleable and ductile.	When a force causes metal ions to move past each other, layers of ions are still held together by the delocalised electrons between them.	

Other properties of metals

Metals generally have a high density. The cations in a metal lattice are closely packed. The density of a metal depends on the mass of the metal ions, their radius and the way in which they are packed in the lattice.

Metals are good conductors of heat. When the delocalised electrons bump into each other and into the metal ions, they transfer energy to their neighbour. Heating a metal gives the ions and electrons more energy and they vibrate more rapidly. The electrons, being free to move, transmit this energy rapidly throughout the lattice.

Metals are lustrous. Because of the presence of free electrons in the lattice, metals reflect light of all wavelengths and appear shiny.

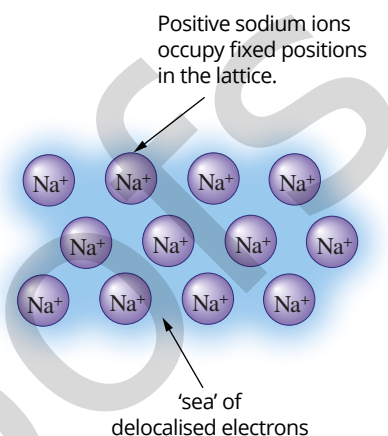


FIGURE 4.2.2 A representation of a sodium metal lattice. Each sodium atom loses its one valence electron. This electron is shared with all atoms in the lattice to form a sea of delocalised electrons.

Metals tend to react by losing electrons. The delocalised electrons in metals may participate in reactions anywhere on the metal's surface. The reactivity of a metal depends on how easily electrons can be removed from its atoms. This is covered in more detail in section 4.3.

Limitations of the metallic bonding model

Although this model of metallic bonding explains many properties of metals, some cannot be explained so simply. These include the:

- range of melting points, hardness and densities of different metals
- differences in electrical conductivities of metals
- magnetic nature of metals such as cobalt, iron and nickel.

In order to deal with such questions, you would need a more complex model of metallic bonding, which is beyond the scope of this book.

Worked example 4.2.1

EXPLAINING THE CONDUCTIVITY OF ALUMINIUM

With reference to the electron configuration of aluminium, explain why solid aluminium can conduct electricity.	
Thinking	Working
Using the atomic number of the element, determine the electron configuration of its atoms. (You may need to refer to a periodic table.)	Al has an atomic number of 13. This means a neutral atom of aluminium has 13 electrons. The electron configuration is 2,8,3.
From the electron configuration, find how many outer-shell electrons are lost to form cations that have a stable, noble gas electron configuration. These electrons become delocalised.	Al has three electrons in its outer shell. Al atoms will tend to lose these three valence electrons to form a cation with a charge of +3. The outer-shell electrons become delocalised and form the sea of delocalised electrons within the metal lattice.
An electric current occurs when there are free-moving charged particles.	If the Al is part of an electric circuit, the delocalised electrons are able to move through the lattice towards a positively charged electrode.

Worked example: Try yourself 4.2.1

EXPLAINING THE CONDUCTIVITY OF MAGNESIUM

With reference to the electron configuration of magnesium, explain why solid magnesium can conduct electricity.

4.2 Review

SUMMARY

- Metallic bonding is the electrostatic force of attraction between a lattice of positive ions and delocalised valence electrons. The lattice of cations is surrounded by a sea of delocalised electrons.
- The metallic bonding model can be used to explain the properties of metals, including their malleability, thermal conductivity, generally high melting point and electrical conductivity.

KEY QUESTIONS

- 1 The properties of calcium mean that it is classed as a metal.
 - a Draw a diagram to represent a calcium metal lattice.
 - b Describe the forces that hold this lattice together.
- 2 Barium is an element in group 2 of the periodic table. It has a melting point of 850°C and conducts electricity in the solid state. Describe how the properties of barium can be explained in terms of its bonding and structure.
- 3 Graphite is a non-metallic substance that can be lustrous and conducts electricity and heat. It is not malleable, but breaks if a force is applied.
 - a What properties does graphite share with metals?
 - b What inferences can you make about the structure of graphite given it shares these properties with metals?

4.3 Reactivity of metals



FIGURE 4.3.1 When water is dropped onto metallic potassium, hydrogen gas is produced.

CHEMFILE

Reactivity of group 1 metals

Group 1 metals are so reactive that they must be handled with great care. They need to be stored under oil to prevent the metal coming into contact with moisture in the atmosphere (Figure 4.3.2).

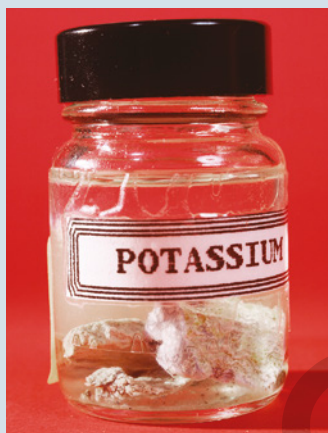


FIGURE 4.3.2 Potassium metal is stored under oil to prevent contact with moisture.

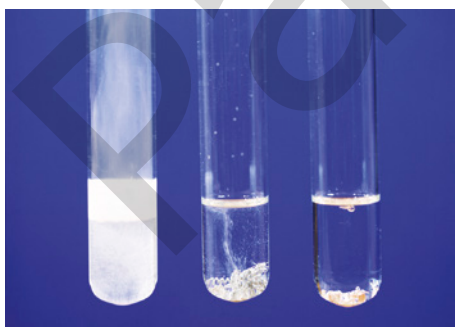


FIGURE 4.3.3 Metals reacting with an equal amount of dilute hydrochloric acid. From left to right: magnesium ribbon, iron filings and copper turnings

In the previous section, you learnt that metals have many common properties. Metallic elements can also have very different properties. These include their **reactivity** with water, acids and oxygen. Some metals are extremely reactive and others are much less so.

This section will look at how the reactivity of different metals can be determined experimentally and explore some of the periodic patterns that exist.

DETERMINING THE REACTIVITY OF METALS

Reactivity with water

The way metals react with water can indicate their relative reactivity.

Figure 4.3.1 shows the reaction of potassium, a group 1 metal, with water. Enough heat is generated to instantly melt the potassium and ignite the hydrogen. The vigour of the reaction is an indication of the reactivity of the metal. Potassium has high reactivity with water, which is characteristic of the group 1 metals.

Table 4.3.1 describes the reaction of some group 1 and group 2 metals with water. In each case, a reaction results in the formation of hydrogen gas.

TABLE 4.3.1 Reaction of selected group 1 and 2 metals with water

Period	Group	Element	Reaction with water
3	1	sodium	reacts vigorously, producing enough energy to melt the sodium, which fizzes and skates on the water surface
4	1	potassium	reacts violently, making crackling sounds as the heat evolved ignites the hydrogen produced by the reaction
5	1	rubidium	Explodes violently on contact with water
3	2	magnesium	will not react with water at room temperature but will react with steam
4	2	calcium	reacts slowly with water at room temperature

From these and other experimental observations we can say that, in general:

- metals in group 1 of the periodic table (i.e. Na, K and Rb) are more reactive in water than those in group 2 (i.e. Mg and Ca)
- going down a group, the reactivity of the metal in water increases.

i The reactivity of metals in water increases down a group and decreases across the period from left to right.

Transition metals

Transition metals are generally less reactive with water than group 1 and 2 metals are. For example, iron reacts fairly slowly with water. Gold and platinum are essentially unreactive.

Reactivity with acids

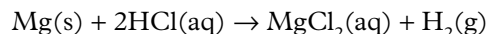
The reactivity of different metals with acids follows the same general patterns as the reactivity of metals with water. Metals are normally more reactive with acids than with water. More metals react with acids and the reactions tend to be more energetic.

Metals can be placed in an order of their relative reactivity. In Figure 4.3.3, the reactions of magnesium, iron and copper with hydrochloric acid are shown. The large amount of bubbling and the mist produced show that magnesium is the most

reactive metal of the three, whereas copper is the least. The bubbles are due to the production of hydrogen gas.

magnesium + hydrochloric acid → magnesium chloride + hydrogen gas

This equation can also be written using chemical formulae:

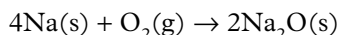


Reactivity with oxygen

Many metals also react with oxygen. The group 1 metals all react rapidly with oxygen. Figure 4.3.4 shows sodium metal burning in a container of pure oxygen. The metal atoms and oxygen molecules rearrange to form a new compound, sodium oxide. The word equation for this reaction is:

sodium + oxygen → sodium oxide

This equation can be written using the chemical formula:



The oxygen in sodium oxide is present as negatively charged oxide ions, O^{2-} . Negatively charged ions are called anions.

The group 2 metals also react with oxygen to form oxides, although not as rapidly as group 1 metals. Heat is usually required to start the reaction.

While the transition metals are less reactive with oxygen than the metals in groups 1 and 2, their reactions are also important. Iron forms rust (iron oxide) when exposed to oxygen and water over a period of time. Many transition metals needed by society cannot be found in nature as pure elements but often exist as oxides.

Iron, copper, titanium and aluminium are all mined as the oxides and must be processed to obtain the finished metal. Figure 4.3.5 shows a cluster of crystals of rutile, the oxide from which titanium is extracted. The production of iron from iron oxide is covered in detail in Section 4.4.

Gold and platinum, which are much less reactive than most other metals, are found in Earth's crust in their pure form. Gold is often found in rock formations called seams alongside quartz, as shown in Figure 4.3.6.



FIGURE 4.3.4 Sodium burning in pure oxygen

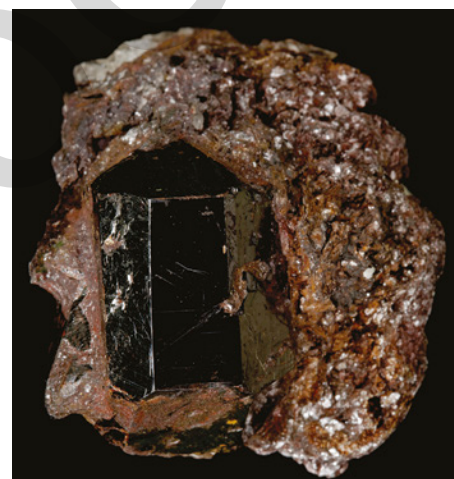


FIGURE 4.3.5 Rutile is a mineral that contains titanium dioxide.



FIGURE 4.3.6 Gold, an unreactive metal, exists in Earth's crust in its metallic elemental form.

REACTIVITY SERIES OF METALS

Chemists have used experimental data from the reactions of metals to produce a **reactivity series of metals**, as shown in Figure 4.3.7. Group 1 metals are at the top of the series, while transition metals appear at the bottom.

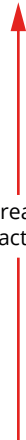
potassium, K	 increasing reactivity
sodium, Na	
calcium, Ca	
magnesium, Mg	
aluminium, Al	
zinc, Zn	
iron, Fe	
lead, Pb	
copper, Cu	
silver, Ag	
gold, Au	
platinum, Pt	

FIGURE 4.3.7 The reactivity series of some common metals

i The larger the metal atoms, the more reactive the metals are. The further the valence electrons are from the nucleus, the more easily they are removed and the more easily they are involved in chemical reactions.

Reasons for different reactivities of metals

In general, the reactivity of main group metals increases going down a group in the periodic table and decreases across a period. This trend in reactivity can be explained in terms of the relative attractions of valence electrons to the nucleus of atoms.

When metals react, their atoms tend to form positive ions by donating one or more of their valence electrons to other atoms. The metal atoms that require less energy to remove electrons tend to be most reactive. The most reactive metals tend to be those with the largest atomic radii and therefore the lowest ionisation energies, which are found in the bottom left-hand corner of the periodic table.

EXTENSION

Extracting iron from iron ore

Modern society is very dependent on iron. About 98% of world iron production is used to make steel. The steel in turn is used in bridges, buildings and all forms of transport. It also has many other uses (Figure 4.3.8).



FIGURE 4.3.8 Steel is used in (a) the Sydney Harbour Bridge, (b) building frames in construction, (c) train tracks and (d) surgical instruments.

Australia is the world's largest exporter of iron ore (a natural compound containing a metal). Australia exported a record 767 million tonnes of iron ore in 2015. Most of the identified deposits of iron ore in Australia—almost 93% (totalling 64 billion tonnes)—are found in Western Australia. Massive deposits of iron ore in the Pilbara region of Western Australia are mined by open-cut methods (Figure 4.3.9).

Iron ore is composed mainly of iron(III) oxide combined with rocky material. The iron must be extracted from the ore before it can be used to make steel. In Australia, the iron oxides in iron ore are usually in the form of haematite (Fe_2O_3).



FIGURE 4.3.9 Iron ore is sourced from open-cut mines in Australia.

The raw materials used in the extraction of iron are:

- iron ore
- coke (a form of carbon)
- limestone
- air.

Extraction of iron from ore and the removal of unwanted materials is carried out in a tall, bottle-shaped tower called a blast furnace. A building containing a modern blast furnace and external conveyer belts is shown in Figure 4.3.10. These furnaces are heated to very high temperatures and are operated continuously for many years.

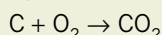


FIGURE 4.3.10 A blast furnace used to extract iron from iron ore. Iron ore is added continuously to the top of the furnace by the conveyor belts on the left-hand side.

Figure 4.3.11 shows a diagram of the inside of a blast furnace. Pre-heated air is blasted into the bottom part of the furnace, while solid ‘charges’ (scoops) of iron ore, coke and limestone are continuously added to the top. The construction of the furnace causes different temperature zones where different reactions can take place.

As the air rises through the furnace and meets the descending charge, oxygen reacts with coke to produce carbon dioxide, which then reacts further with the coke to produce carbon monoxide. These reactions are represented by the following equations:

carbon + oxygen → carbon dioxide



carbon dioxide + carbon → carbon monoxide

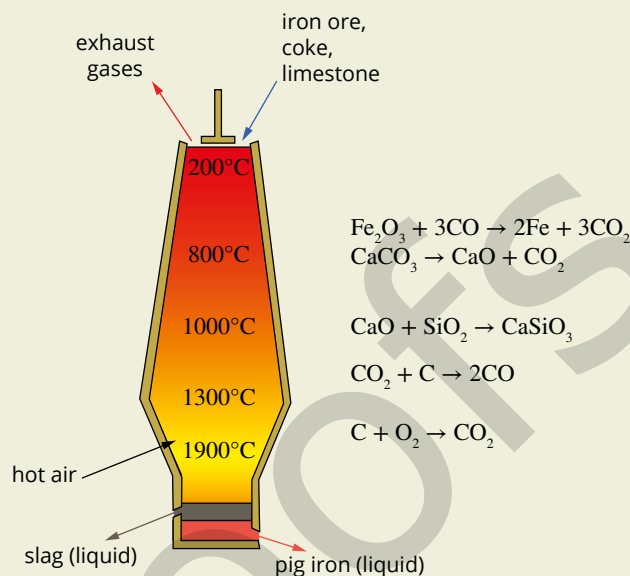
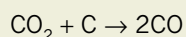
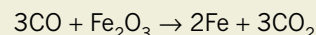


FIGURE 4.3.11 A diagram of a blast furnace. Hot air jets blast air in at the bottom. Iron ore, coke and limestone are added at the top.

The first reaction releases considerable heat energy and helps to maintain the high temperature that provides the fast rate of reaction required for metal production in the furnace. No external source of heat is needed to achieve temperatures up to 1800°C within the furnace.

Extraction of iron from iron oxide occurs in a series of steps in which carbon monoxide is the main reactant. The steps occur in different temperature zones within the blast furnace. The process is summarised by the single chemical equation:

carbon monoxide + iron oxide → iron + carbon dioxide



Iron ore is not pure iron oxide. It also contains rocky material such as silica (silicon dioxide), alumina and manganese oxides. These would not melt in the heat of the furnace and would eventually clog it up.

The limestone (CaCO_3) added to the top of the furnace breaks down in the furnace to form calcium oxide (CaO) and carbon dioxide. Calcium oxide can react with the unwanted materials to form a new compound called slag. Most of the slag is calcium silicate (CaSiO_3) due to the high amount of silica in iron ore.

Both the iron and slag are molten (melted) and sink to the bottom of the furnace. The slag is less dense and floats on top of the molten iron. This provides a cover that prevents the iron from reacting with the incoming air and re-forming iron oxide.

Holes at the base of the furnace are opened and the molten iron and slag are drained out and separated. In a steel works, the iron is usually transferred, while still molten, directly to a steel-making furnace. The slag may be used as a road-surfacing material or to manufacture cement.

4.3 Review

SUMMARY

- The reactivity of different metals with water, acid and oxygen can be determined experimentally.
- Group 1 metals are very reactive with water.
- Metals tend to be more reactive with acids than with water.
- The experimental results of metals reacting with acids are used to place metals in an order of reactivity called the reactivity series.
- The most reactive metals are found at the bottom left-hand side of the periodic table.
- Although the reactivities of transition metals vary, they are usually less reactive than the main group metals.

KEY QUESTIONS

- 1 Predict the products of the reaction between potassium and water and write a word equation for this reaction.
- 2
 - a Describe the trend in reactivity with water down the group 1 metals.
 - b What atomic property accounts for this trend in reactivity?
- 3 Pieces of iron, zinc and gold metal were each placed into test-tubes containing hydrochloric acid. The table below summarises the observations from each test. On the basis of the reactions, list the metals in order of reactivity.
- 4 Using the reactivity series of some common metals (Figure 4.3.7 on page xxx), determine whether calcium, platinum or aluminium would react most vigorously with oxygen in the air.

Metal	Evidence of reaction
iron	bubbles of gas are slowly produced
	iron metal disappears after a very long time
zinc	bubbles of gas are rapidly produced
	zinc metal disappears
gold	no signs of a chemical reaction
	gold does not change appearance

4.4 Modifying metals

The way in which a metal can be used is determined by the metal's physical and chemical properties. Although some metals are valuable in their pure state, most metals need to be modified to make them more useful.

Modified metals have a wide range of applications. In Figure 4.4.1, you can see a number of uses of modified metals. In each case, a metal with desirable properties has been chosen and improved by the processes covered in this section.

A metal can be modified in different ways, including:

- through alloy production
- by heat treatment
- formation of nano-sized structures.

In this section, you will look at the different ways metals are modified and the effect of these modifications on the metal's properties and uses.

MAKING ALLOYS

Often, metals are mixed with small amounts of another substance, usually a metal or carbon. The substances are melted together, mixed and then allowed to cool. The resultant solid is an alloy.

By varying the composition of alloys, you can obtain materials with specific properties. Generally, an alloy is harder and melts at a lower temperature than the pure metal. This is because atoms of different sizes are now included in the metal lattice. As these atoms do not pack in the same way as the main metal, they will not allow the lattice to shift and bend in the same way. This disruption of the regular metallic lattice also accounts for the lowered melting point.

Steel: alloys of iron

Currently, almost all of the iron mined around the world is used to make the alloy steel. The simplest steel is made by adding a small amount of carbon to iron to make carbon steel.

Carbon steel is a type of **interstitial alloy**. In interstitial alloys, a small proportion of an element with significantly smaller atoms is added to a metal. The added atoms sit in interstices (very small spaces) between metal cations in the metallic lattice, as shown in Figure 4.4.2.

Carbon steel is generally harder and less malleable than pure iron. Varying the amount of carbon in the mixture produces steels with different properties, as shown in Table 4.4.1. This allows the steel with the best properties to be used in specific applications.

TABLE 4.4.1 A summary of the properties and uses of some different carbon steels

Type of carbon steel	Percentage of carbon	Properties	Typical use
low carbon steel	less than 0.3%	strong, easily shaped	bridges, buildings, ships and vehicles
medium carbon steel	0.3–0.45%	increased hardness and tensile strength, decreased ductility	large machinery parts
high carbon steel	0.45–0.75%	very strong, more brittle	springs and high-strength wires
very high carbon steel	up to 2.5%	hard, more brittle	cutting tools

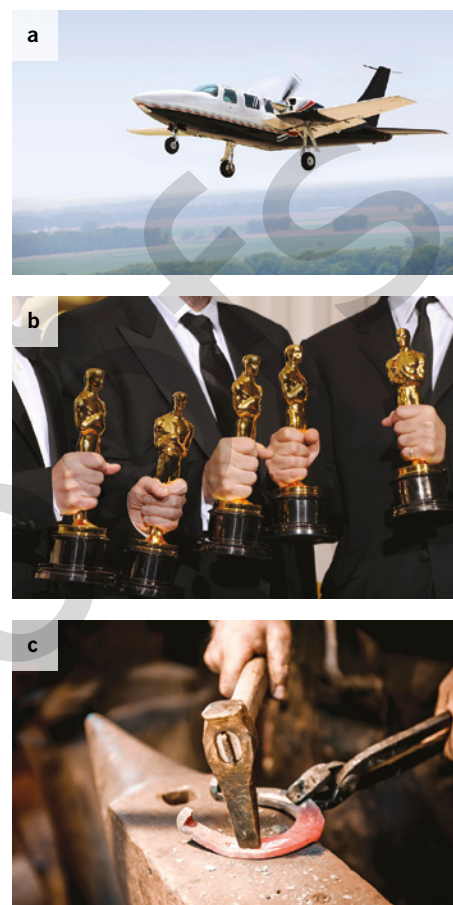
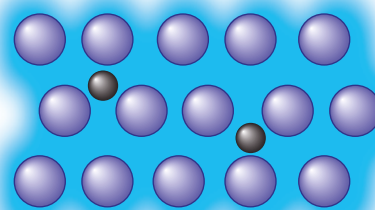


FIGURE 4.4.1 (a) Modified metals are used in aeroplanes, which need to be made from materials that are strong, durable and light. (b) The Oscar statuettes are made of britannium (a mixture of tin, copper and antimony) and then coated successively in layers of copper, nickel, silver and gold. (c) Blacksmiths use heat and hammering tools to modify the properties of the metal used for horseshoes.



The smaller carbon atoms occupy some of the spaces between the iron atoms.

FIGURE 4.4.2 Steel is an interstitial alloy of iron and carbon. Note the relative sizes and position of the atoms in an interstitial alloy.

Other elements can also be added in addition to carbon to make steels with improvements to different properties. Examples of some of these are listed in Table 4.4.2.

TABLE 4.4.2 Some elements that are alloyed with iron to make steels

Alloying element	General effects on properties	Example
manganese	increases strength and toughness	bicycle frames
chromium	- increases hardness and tensile strength - resists corrosion	stainless steel for cutlery, kitchen sinks
nickel	- increases toughness, tensile strength and hardness - resists corrosion	stainless steel
cobalt	- improves magnetic properties - resists high temperatures	alnico magnets, jet propulsion engines

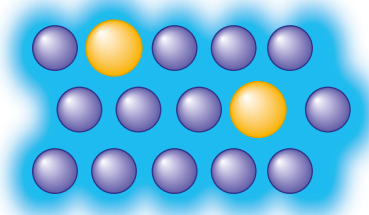


FIGURE 4.4.3 The iron, nickel and chromium atoms in stainless steel are relatively similar in size. The alloy is harder than any of these metals alone.

The steels in Table 4.4.2 are not considered to be interstitial alloys, as the added atoms are too big to fit into the spaces in the metallic lattice. The atoms of the elements added replace some of the iron cations and the mixture is called a **substitutional alloy**. In general, the metallic elements added to make substitutional alloys have fairly similar chemical properties and form cations of a similar size to the main metal.

Stainless steel is a substitutional alloy of nickel and chromium. The nickel and chromium atoms take the place of some of the iron atoms in the lattice (Figure 4.4.3). All the metal cations are attracted to the sea of electrons, so the lattice is still strongly bonded. However, because the different kinds of atoms are slightly different in size, the layers within the lattice cannot move as easily past each other. This makes the alloy harder and less malleable than pure iron.

i When other metals or carbon are added to melted metals, the substance that is produced is called an alloy. Alloys tend to have improved properties compared to the original metal, making them more useful in society.

Common alloys

Many common materials are made of alloys in order to make the metal more suited to a particular purpose. Metals are blended together to combine properties, such as strength, colour, reflectivity and chemical stability. The resulting alloy will have different properties from the individual elements. A number of examples of substitutional alloys are shown in Table 4.4.3.

WORK HARDENING AND HEAT TREATMENT

The way a metal object is prepared also affects how it behaves. Many metals are prepared in the liquid state and then cooled. The rate at which a metal is cooled affects the properties of the solid.

The model that you have been developing for the structure of metals describes the arrangement of particles within a single metal **crystal**. A crystal is a region in a solid where the particles are arranged in a regular way. A sample of solid metal consists of many small crystals. Each crystal is a continuous regular arrangement of cations surrounded by a sea of delocalised electrons, but the arrangement of individual crystals with respect to one another is random like those shown in Figure 4.4.5. As you can see in Figure 4.4.5, the regular lattice is disrupted at the point where one crystal meets another.

CHEMFILE

Gold

Pure gold is 24-carat gold. It is so soft that it can be moulded by hand. Gold in jewellery is a substitutional alloy (Figure 4.4.4). Nine-carat gold is 9/24 gold, which is 37.5% gold and 62.5% other metals. White gold is an alloy of gold and a metal such as nickel, manganese or palladium. The purity of white gold is also expressed in carats.

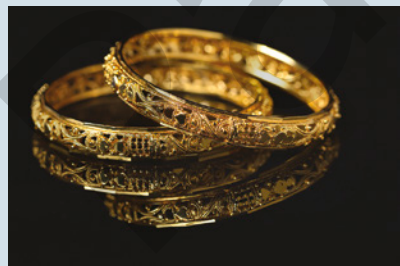


FIGURE 4.4.4 Gold jewellery is typically made from alloys. Pure gold (24-carat purity) is not often used for jewellery because it is too soft.

TABLE 4.4.3 Substitutional alloys found in everyday objects

Alloy	Metals used	
Australian 20-cent coin	copper (75%) and nickel (25%)	
Australian \$2 coin	copper (92%), aluminium (6%) and nickel (2%)	
dental mercury amalgam	mercury and zinc	
brass	copper (65%) and zinc (35%)	
bronze	copper (90%) and tin (10%)	

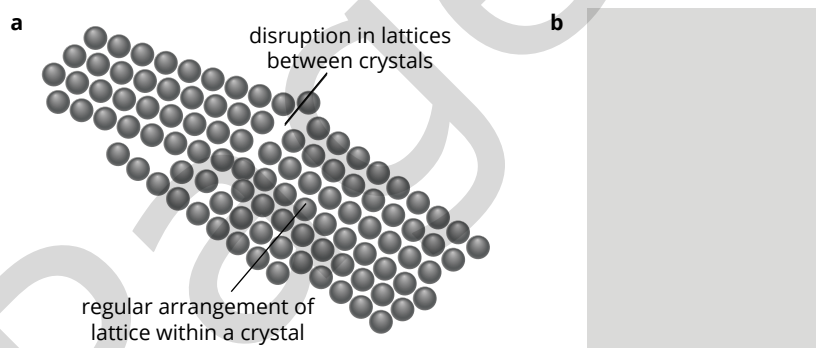


FIGURE 4.4.5 (a) Metals are made of a number of crystals, which have boundaries between them. (b) An electron micrograph shows the individual crystals that formed as a molten metal mixture cooled down and solidified.

The way a metal behaves—its malleability and brittleness—will depend on the size and arrangement of the crystals. Generally, smaller crystals result in harder metals as there is less free movement of layers of cations over each other. Smaller crystals also have more areas of disruption between them, and this usually means that these metals will be more brittle.

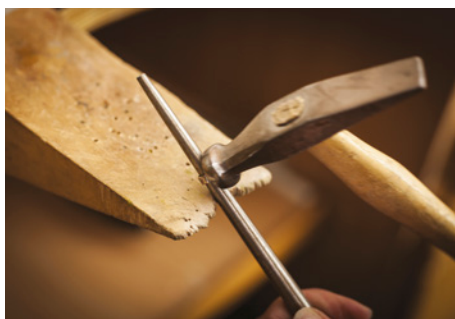


FIGURE 4.4.6 The crystal structure of metals like this gold ring can be altered by work hardening. Care is needed as the gold becomes more brittle through this process.

CHEMFILE

Work hardening with paperclips

You can see work hardening by bending a paperclip. If you bend it once, it remains fairly pliable, but if you bend it backwards and forwards several times, it snaps. Bending rearranges the crystal grains, making the metal harder but more brittle.

The crystal structure of metals can be altered in a number of ways. Two of these ways are **work hardening** and **heat treatment**.

Work hardening

Hammering or working cold metals causes the crystals to rearrange as they are pushed and deformed. This can result in the hardening of the metal as the crystals are flattened out and pushed closer together. Figure 4.4.6 shows a gold ring being hammered on a round tool, called a triblet, to harden and strengthen it.

Heat treatment

The physical properties of a metal can be altered by controlled heating and cooling. The three main methods of heat treatment (**annealing**, **quenching** and **tempering**) of metals and the effects on the metal's properties and structure are summarised in Table 4.4.4.

TABLE 4.4.4 Summary of heat treatment methods and their effect on metal properties

Treatment	Process	Effect on metal structure	Effect on metal properties
annealing	A metal is heated to a moderate temperature and allowed to cool slowly.	Larger metal crystals form.	The metal is softer with improved ductility.
quenching	A metal is heated to a moderate temperature and cooled quickly (sometimes by plunging into water).	Tiny metal crystals form.	The metal is harder and brittle.
tempering	A quenched metal is heated (to a lower temperature than is used for quenching) and allowed to cool.	Crystals of intermediate size form.	The metal is hard but less brittle.

When metals are heated above a critical temperature, the individual crystals merge. When the metal is allowed to cool, the crystals re-form. The rate of cooling determines how large the new crystals will be. Faster cooling leads to smaller crystals; slower cooling allows more time for crystals to grow larger.

Steels respond well to heat treatment. Annealing steels reduces their strength or hardness, increases uniformity of crystals and so reduces stresses and restores ductility. Hardening of steels by quenching and tempering increases the strength and wear properties (Figure 4.4.7).

CHEMISTRY IN ACTION

Swords

The art of producing a superior sword blade was a highly valued skill in ancient Japan. It required knowledge of the properties of alloys and the effects of heat-treating and work hardening. The blade was heated and plunged into cold water to harden the edge.

To avoid the blade becoming too brittle, the sword was then tempered. The wavy line along the blade shows the interface between hard and flexible regions within the metal.



FIGURE 4.4.7 Japanese swordsmiths had to understand the properties of alloys and the effects of heat-treating and work hardening.

METALLIC NANOMATERIALS

Metallic nanomaterials consist of metal atoms arranged to make structures which has a least one dimension between 1–100 nm ($1\text{ nm} = 10^{-9}\text{ m}$). Metallic nanomaterials have very different properties when compared to a bulk sample of the same type of metal. Scientists have long been fascinated by the effects of nanomaterials. For example, gold–silver alloyed nanoparticles change the colour of glass (Figure 4.4.8).

As you learnt in Section 1.2 on page xxx, emerging technologies have allowed scientists to manipulate metals at the nanoscale and create different forms of metallic nanomaterials.

Forms of metallic nanomaterials

Materials exist in many different forms at the nanoscale level, just as they do on the macroscale level. These forms include:

- particles
- rods
- wires
- tubes.

Metallic nanoparticles

Nanoparticles are usually spherical particles that range from about 1 to 100 nm in size. They have a very high surface area to volume ratio compared to the bulk material of similar volume.

The gold nugget shown in Figure 4.4.9 consists of many billions of cations and delocalised electrons. The properties of gold and other metals, such as electrical and thermal conductivity and metallic lustre, are explained in terms of the movement of delocalised electrons.

However, scientists have found that in a metal crystal with less than a certain number of atoms, the number of valence electrons released is too small to behave as a ‘sea’ of delocalised electrons. Metal nanoparticles that contain only 100 or so atoms have properties in-between those of metals and non-metals. In general, the nanoparticles have different optical properties and are more sensitive to heat.

For example, gold takes on a ruby colour when the gold particles are reduced to nano-size. You can see this in the stained glass window in Figure 4.4.8. The deep yellow colour is caused by gold–silver alloyed nanoparticles.

Metallic nanorods

Nanorods are nanoscale rods in which each dimension ranges from 1 to 100 nm. They have a length to width ratio of 3:1–5:1 (Figure 4.4.10). Nanorods of metals such as gold and silver have been synthesised in the laboratory. Their applications are diverse, including in display technologies and microelectronics, powering everything from solar cells to mobile phones.

Metallic nanowires

A **nanowire** is a nano-sized wire. The diameter of a nanowire is measured on the nanoscale, but its length is unrestricted. Nanowires differ from nanorods because they are much longer, as shown in Figure 4.4.11. Platinum nanowires have promising applications as catalysts and in electronics.

Uses of metallic nanomaterials

Nanomaterials have unique electrical, catalytic, magnetic, mechanical, thermal and imaging characteristics. This makes them attractive for use in medical, pharmaceutical, electronic and different engineering sectors.



FIGURE 4.4.8 As light passes through this stained glass window containing gold–silver alloyed nanoparticles, it causes the glass to change colour.



FIGURE 4.4.9 Bulk gold has a characteristic colour and typical metallic properties.



FIGURE 4.4.10 A group of nanorods

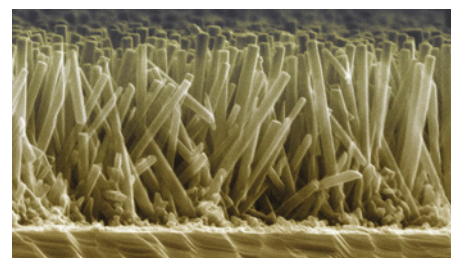


FIGURE 4.4.11 Zinc oxide nanowires

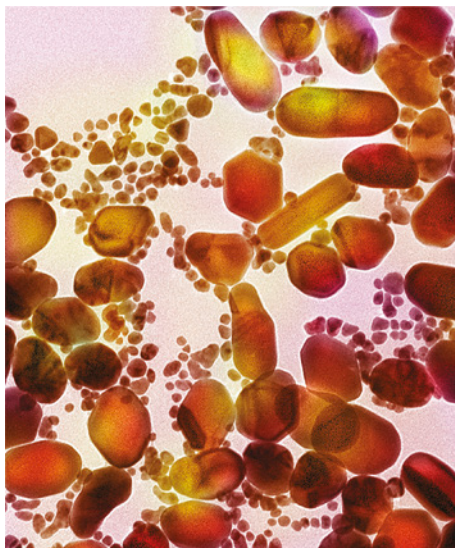


FIGURE 4.4.12 Applications of gold nanoparticles include cancer treatment research and detecting biological toxins.



FIGURE 4.4.13 A wound dressing embedded with silver nanoparticles can be used to kill bacteria.



FIGURE 4.4.14 A nanotechnology copper-based solder has advantages in situations where long life and reliability are critical.

Gold nanoparticles in cancer research

Gold nanoparticles are the subject of substantial research with a wide range of applications (Figure 4.4.12). One area of development is in using gold nanoparticles as a targeted chemotherapy treatment method.

Gold nanoparticles can be attached to molecules of a tumour-killing agent known as tumour necrosis factor (TNF). The nanoparticles hide the molecule from the body's immune system.

The nanoparticles carrying the TNF tend to accumulate in cancer tumours, allowing TNF to destroy tumours. The nanoparticles do not appear to accumulate in other regions of the body, which means healthy cells are not affected.

Silver nanoparticles kill bacteria

Silver ions have long been known to kill bacteria. The ions can rapidly penetrate bacterial membranes and interact with proteins in the bacteria, destroying the cell structure of the bacteria and preventing them from reproducing.

Technology has enabled silver nanoparticles to be included in many different types of wound dressings. When the dressing (Figure 4.4.13) comes into contact with moisture from the wound, silver nanoparticles are slowly but continuously released from the wound pad. They then enter the wound and kill bacteria.

In similar antibacterial applications, Samsung has created and marketed a material called Silver Nano™, which adds silver nanoparticles to the surfaces of household appliances. Silver nanoparticles have been embedded in the surfaces of plastic storage bins, as well as in fabrics used by astronauts, babies and outdoor enthusiasts.

Copper nanoparticles go to space

Solder is a filler metal used to join two or more metals. Solders are essential to plumbing and metal constructions, including in satellites and spacecraft. For most of history, solders have contained a high amount of lead. Concerns about the toxicity of lead have driven the development of lead-free solder.

The complex electronics in satellites, such as the solar-powered satellite in Figure 4.4.14 must be reliable and efficient over a very long time. Space scientists have developed a nanotechnology copper-based solder that offers far superior performance over the materials currently in use. It is expected that the new solder material will produce joins with up to 10 times the electrical and thermal conductivity of current solders, with a wide range of space and defence applications.

Iron nanoparticles remove pollution in ground water

Iron nanoparticles are being used to clean up the pollutant carbon tetrachloride (CCl_4) from ground water (Figure 4.4.15). Carbon tetrachloride is a manufactured toxic chemical that has been shown to cause cancer in animals. Spills of carbon tetrachloride can spread through soil and create large areas of contamination.

Iron nanoparticles can quickly and effectively break down carbon tetrachloride to a mixture of relatively harmless products. It may be possible to inject the nano-sized iron deep into the ground where it can treat contaminated ground water.



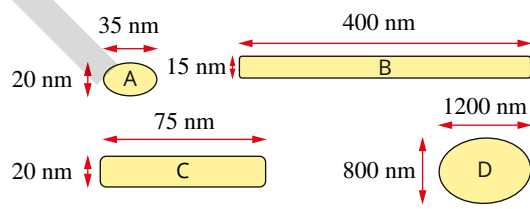
FIGURE 4.4.15 Nanoparticles of iron are being investigated as a way of eliminating a range of environmental pollutants.

4.4 Review

SUMMARY

- Most metals can be modified to alter their properties and make them more useful.
- An alloy is a mixture of a metallic element with other elements.
- Alloys are often harder and melt at a lower temperature than pure metals.
- Heat treatment changes the size and arrangement of crystals in metals.
- Metallic nanomaterials are nano-sized particles of metals that range in size between 1 and 100 nm, where $1 \text{ nm} = 10^{-9} \text{ m}$.
- Different forms of metallic nanomaterials include nanoparticles, nanorods and nanowires.
- Metallic nanomaterials have applications in medical, pharmaceutical, electronic, environmental and engineering sectors.

KEY QUESTIONS

- 1 Consider the alloys that make up a 20-cent coin and high-carbon steel.
 - a List the elements present.
 - b Draw a diagram to show how the atoms of these elements are arranged in the alloy.
- 2 Dentists fit partial dentures by means of small metal hooks. The hooks attach the denture to the remaining teeth. The hooks are easily bent at first to fit snugly in individual mouths. However, if the hooks are bent backwards and forwards too often, then they become brittle and snap. Explain why this happens.
- 3 Before shaping aluminium into objects by hammering, it is usually annealed to make it more malleable. The final product of this shaping process is then usually quenched to increase its strength. Use the metallic bonding model to explain why:
 - a annealing makes aluminium more malleable
 - b quenching can increase the strength of the final aluminium object.
- 4 During the manufacture of a metal object, many different modifications may be needed. Propose a series of modifications to produce a steel chisel that has a flexible and strong shaft with a hard blade.
- 5 Classify the metallic nanoparticles shown in the illustration.

Particle A: 35 nm diameter sphere.
Particle B: 400 nm length, 15 nm diameter rod.
Particle C: 75 nm length, 20 nm diameter rod.
Particle D: 1200 nm diameter sphere.
- 6 Iron nanoparticles must be stored away from oxygen as they will react quickly to produce iron oxide. Write a word equation for the reaction of iron nanoparticles with oxygen.

Chapter review

KEY TERMS

alkali metal	interstitial alloy	quenching
annealing	lattice	reactivity
brittle	malleable	reactivity series of metals
conductor	metallic bonding	steel
crystal	metallic bonding model	substitutional alloy
delocalise	metallic nanomaterials	tempering
delocalised electron	molten	tensile strength
ductile	nanorod	work hardening
heat treatment	nanowire	

Properties of metals

- Which of the following metals would have similar properties to beryllium?
Ca, Cs, Cu, Pb, Mg, Zn, Sr, K
- Use the data in Table 4.1.2 on page xx to answer the following questions.
 - Which metal is the best conductor of heat?
 - Why is this metal not used in saucepans?
 - What metals are used to make saucepans?
- Which property most clearly distinguishes the metals from the non-metals listed in Table 4.1.2 on page xx?
- What do you think is the most important property of each of the following metals that has led to its widespread use?
 - aluminium
 - copper
 - iron
- The atomic number of calcium is 20. How many electrons are in an atom of calcium and in a Ca^{2+} cation?
- Determine the electron configuration of an aluminium atom and the configuration of its most stable cation.
- What is the meaning of the term 'ductile' when referring to metals?

Metallic bonding

- Use the metallic bonding model to explain each of the following observations.
 - Copper wire conducts electricity.
 - A metal spoon used to stir a boiling mixture becomes too hot to hold.
 - Iron has a high melting point, 1540°C .
 - Lead has a density of 11.4 g mL^{-1} , which is much higher than for a non-metal such as sulfur.
 - Copper can be drawn out to form a wire.

- Consider the metallic bonding model used to describe the structure and bonding of metals.
 - What is meant by the following terms?
 - delocalised electrons
 - a lattice of cations
 - metallic bonding
 - Which electrons are delocalised in a metal?
- Describe the arrangement of particles in a metal wire and how they allow the wire to conduct electricity.
- Use a diagram to describe what is meant by the term 'metallic lattice'.

Reactivity of metals

- Look at the periodic table at the end of the book.
 - Name a metal that would have similar properties to calcium.
 - In which part of the periodic table are magnetic metals found?
- Which of the following metals would you expect to be the least reactive with water?
aluminium, sodium, rubidium
- When a reactive metal is added to water, bubbles or fizzing can be observed. Explain the appearance of the bubbles.
- The image to the right shows similar-sized pieces of iron and silver in test-tubes of sulfuric acid of the same concentration. Describe the reactivity of the two metals and identify which metal is on the left and which is on the right.



- 16** Observations of the reaction of metals A, B and C with dilute hydrochloric acid are summarised below. Identify which metal is aluminium, copper and sodium.
- Metal A does not react with dilute hydrochloric acid.
 - Metal B is stored in oil because of its high reactivity, and it undergoes a violent reaction with dilute hydrochloric acid.
 - Metal C reacts very slowly with dilute hydrochloric acid.
- 17** Identify the following statements about the reactions of metals as true or false.
- All metals react with acid to produce hydrogen gas.
 - Hydrogen is a flammable gas.
 - The most reactive metals are located at the top of a group in the periodic table.
 - A metal that is very reactive with water is likely to be less reactive with a solution of a dilute acid.
 - When metals react with oxygen in the air, they release energy.
- 18** Write the word equation for the reaction that occurs when magnesium is heated strongly in air.

Modifying metals

- 19** Alloys are modified metals. In each example, state the metals used and one property that is different from the original metals.
- Australian one dollar coin
 - solder
 - 9-carat gold
 - stainless steel
 - dental mercury amalgam
- 20** A student wishes to make an iron needle more malleable. She heats three needles strongly in a Bunsen burner flame. She then treats each needle differently.
- Needle 1 is allowed to cool slowly on the bench.
 - Needle 2 is cooled quickly by dropping it into a beaker of cold water.
 - Needle 3 is cooled quickly in the beaker of water then reheated in the flame before being allowed to cool more slowly on the bench.
- Place the resulting needles in order of least malleable to most malleable.
- 21** Horseshoes are often made from steel that is worked into shape by a process of heating and hammering the metal. Explain how this process results in a better horseshoe than one simply made of iron.
- 22** A scientist synthesises a new material in the laboratory. It is a solid substance with a diameter of 15 nm and a length of 2300 nm. Which form of nanomaterial is this substance?
- 23** A scientist synthesised a tiny particle for use in solar cells, with a diameter of 8.34×10^{-7} m. Would this particle be classified as a nanomaterial?

- 24** Explain why a nanowire can be longer than 3000 nm but still be considered a metallic nanomaterial.

Connecting the main ideas

- 25** Metals have many uses in modern society.
- Name one metal that is used in large quantities in the building industry.
 - What properties make it suitable for this use?
 - What properties limit its use in buildings?
 - Name one metal that is used in large quantities for making electrical wires.
 - What properties make it suitable for use in wires?
 - Name two metals that are used in large quantities in the jewellery trade.
 - What properties make them suitable for this use?
- 26** Draw a concept map using the following terms: metal, cation, delocalised electron, lattice, electrostatic attraction.
- 27** Some metals are found as elements in nature; others are found as compounds, combined with other elements such as oxygen and sulfur in ores. Australia has natural reserves of many metals and ores, with mining producing large quantities of metals, including aluminium, copper, gold, iron and silver.
- Give the chemical symbol for each of these metals.
 - In which group, period and block of the periodic table are each of these metals found?
 - Which of these metals are found in nature as elements rather than compounds?
 - Which of these metals are transition elements?
 - Which of these elements is the rarest?
- 28** It has been shown using X-rays that metals have a crystalline structure in the solid state.
- Explain, using the metallic bonding model, why metals form crystals.
 - What problems can this crystalline structure of metals cause?
- 29** The boiling points of three metals—sodium, potassium and calcium—are given in the table below.
- | Metal | Boiling point (°C) |
|-------|--------------------|
| Na | 892 |
| K | 760 |
| Ca | 1490 |
- In which group and period of the periodic table are these metals found?
 - Write the electron configuration for each of the three metals.
 - Use the metallic bonding model to suggest why:
 - sodium has a higher boiling point than potassium
 - calcium has a much higher boiling point than potassium.
- 30** Aluminium is the most abundant metal in the Earth's crust. Why was aluminium not available before 1886?

